

Fluorescence & Labeling Strategies

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Fluorescence & Labeling Strategies

- *Basis of fluorescence & Properties of fluorophores*
- *Fluorescent proteins*
- *Organic dyes*
- *Fluorophores photophysics*
- *Protein labeling in live cells (FPs, Self-labeling tags, UAA)*
- *Labeling in fixed cells (Immunostaining, non-specific stains (mb, nucleic acids, etc...), FISH)*



Fluorescence & Labeling Strategies

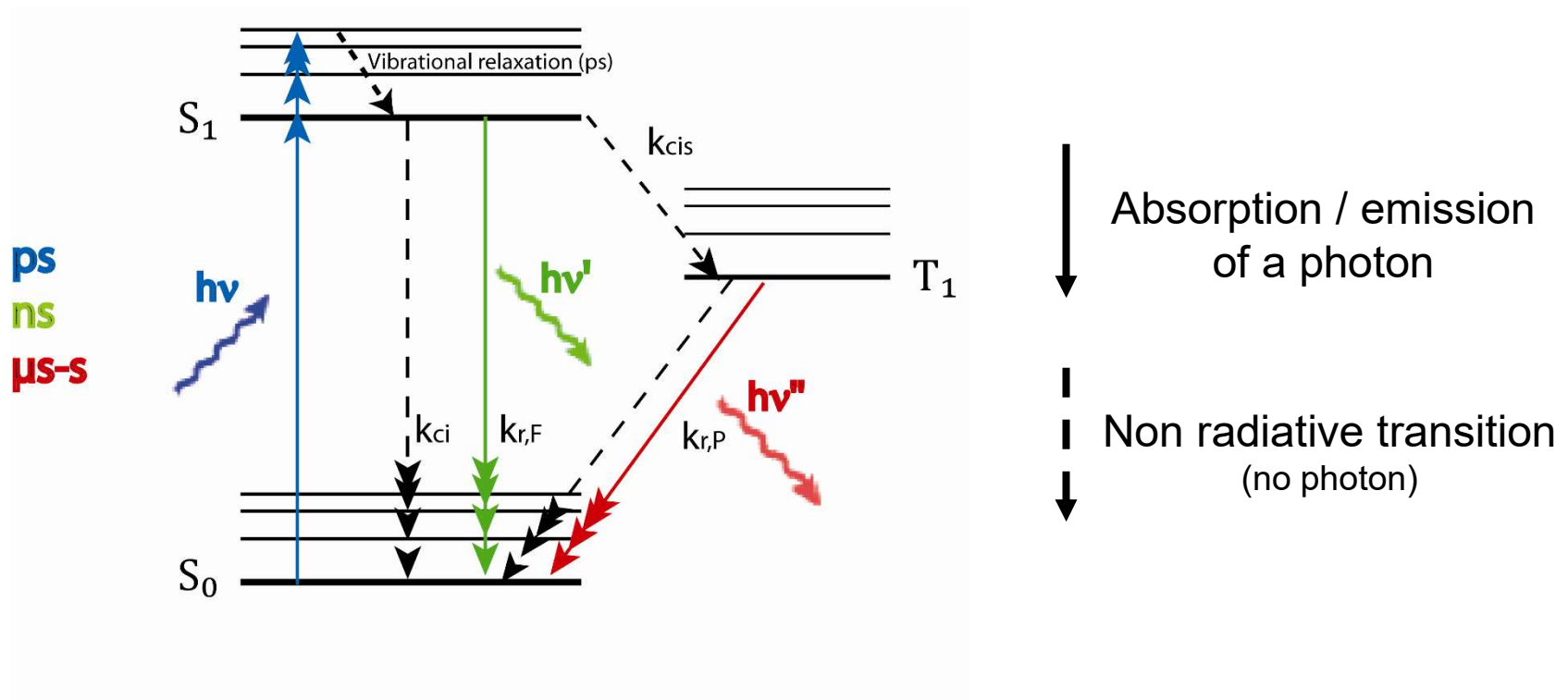
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Fluorescence basics

Fluorophore : chromophore that can emit a photon (through fluorescence(transition $S_1 \rightarrow S_0$)), after an electronic excitation

Simplified Jablonski diagram



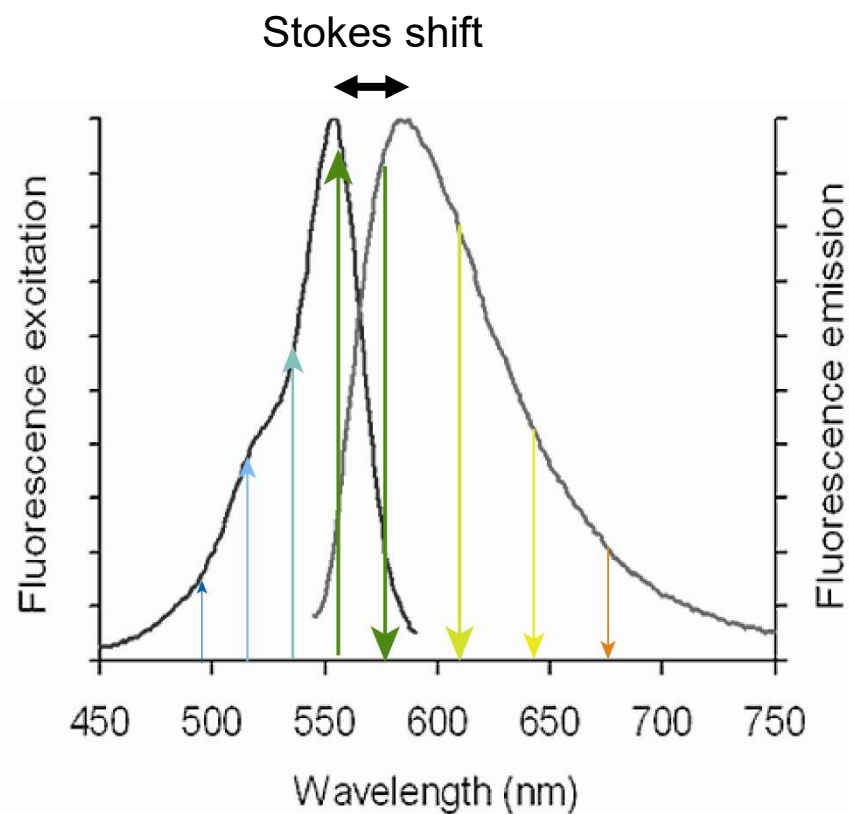
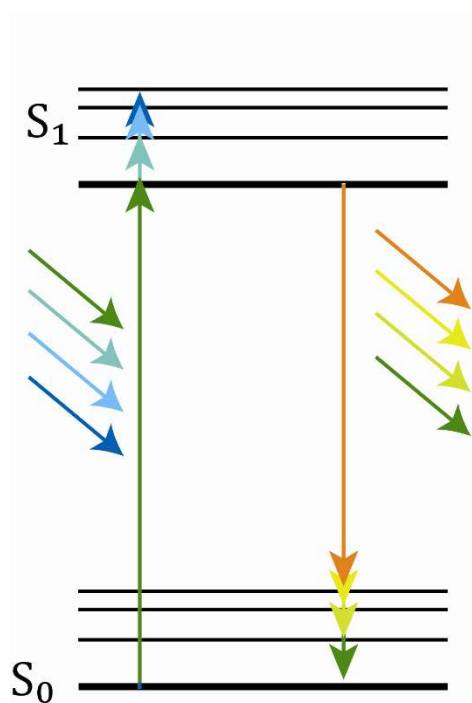
Fluorescence basics

A fluorophore is highly sensitive to its local environment.
Many parameters can influence the properties of these transitions,
leading to many different useful observables

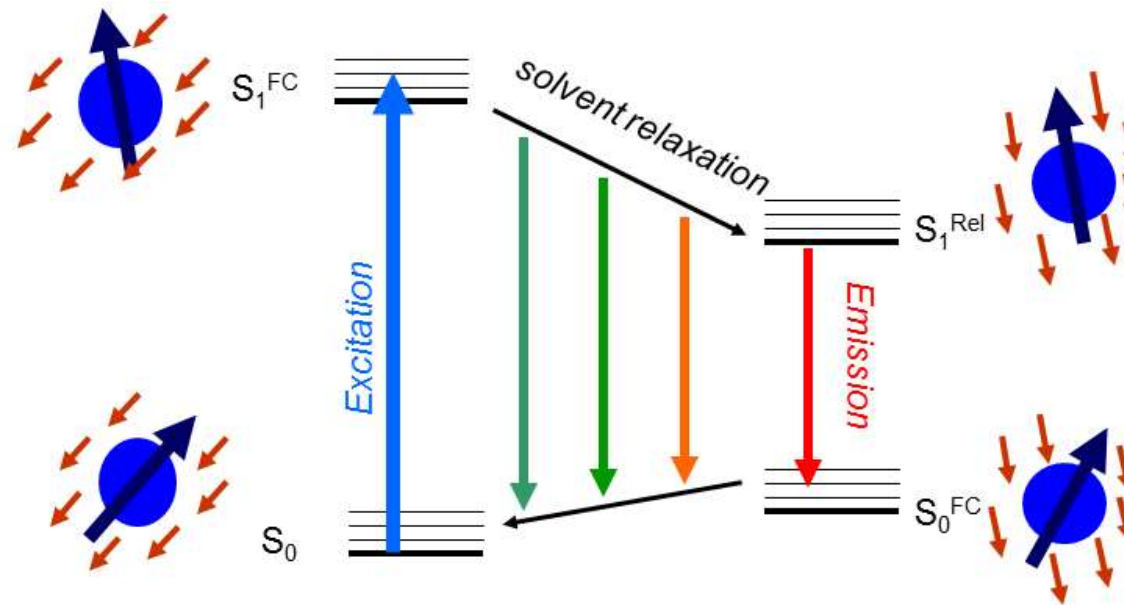
- *Color of the emission*
- *Efficiency of the emission (Quantum Yield)*
- *Kinetics of the emission process (Lifetime)*
- *Resonant energy transfer*
- *Quenching*
- *Polarization of the emission*
- *Change in the fluorophore structure*



Properties : excitation and emission spectra



Effect of the local environment on the color of the emission



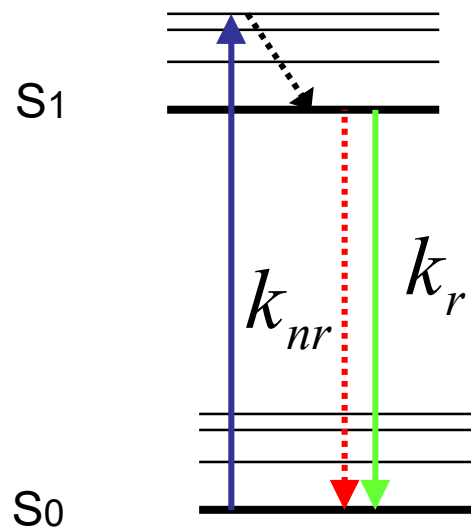
- Solvent polarity and viscosity
- Rate of solvent relaxation
- Nature of the local environment
- Rigidity of the local environment

Examples :

- ***color of the tryptophan fluorescence in proteins***
- ***Color of fluorescent lipids depending on the lipid phase***



Efficiency of the emission



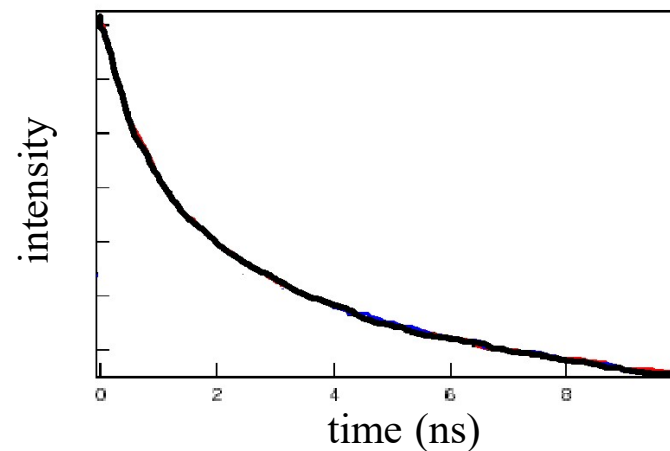
Quantum Yield

$$QY = \frac{k_r}{k_r + k_{nr}}$$

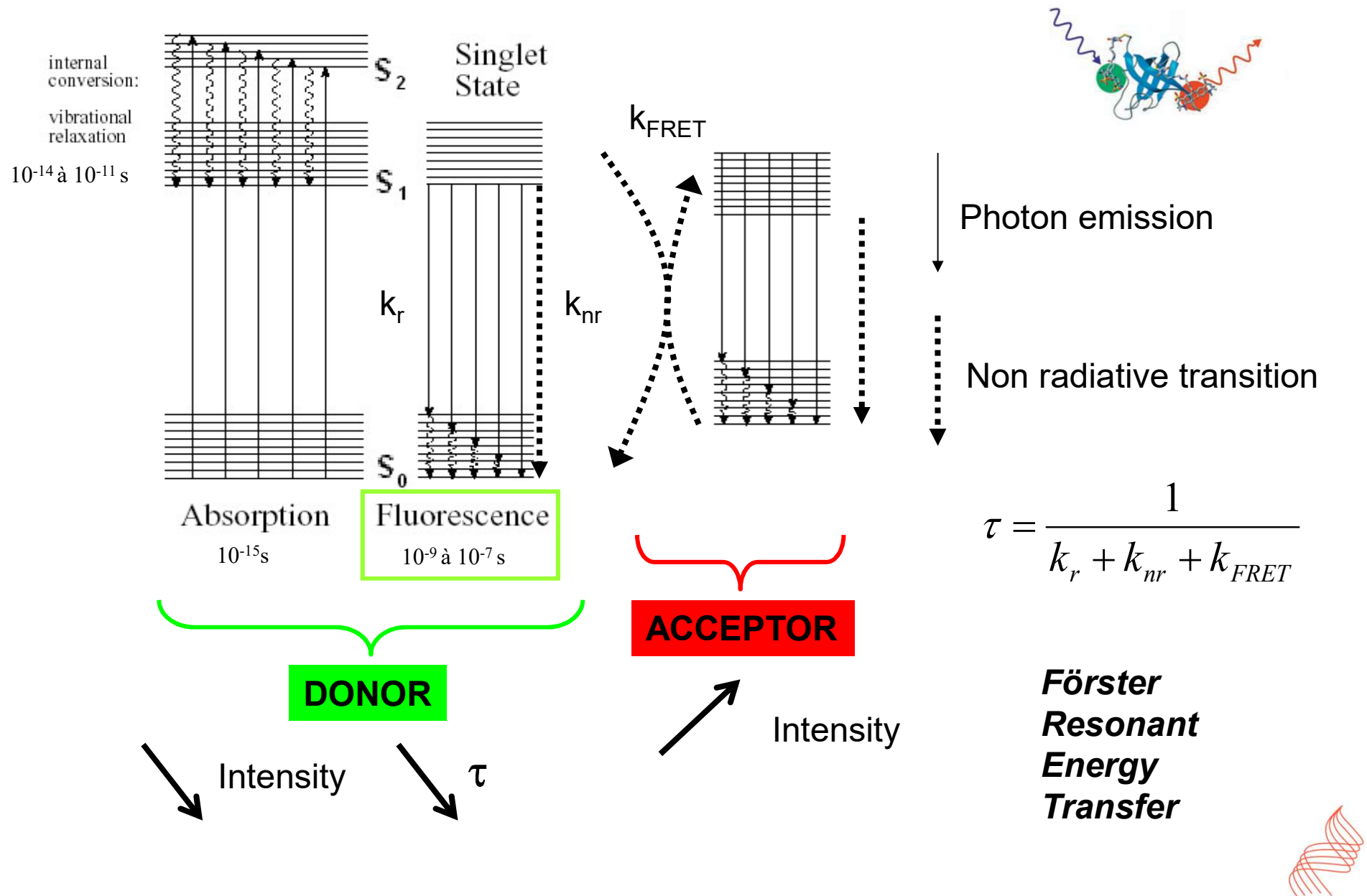
Excited state lifetime

$$\tau = \frac{1}{k_r + k_{nr}}$$

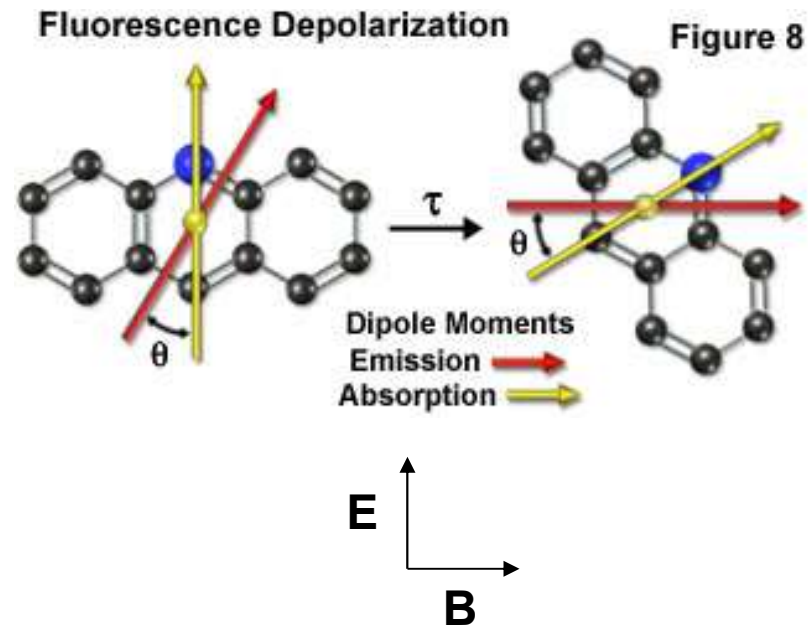
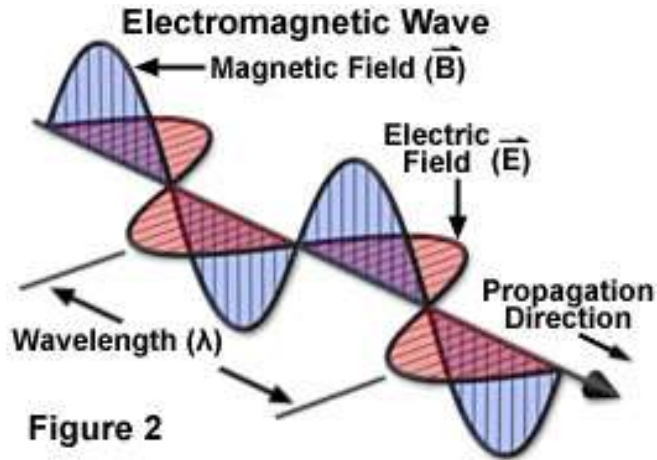
Excited state lifetime can be measured with pulsed excitation and time resolved detection :
time-resolved fluorescence



Excited-state energy transfer to another fluorophore



Polarisation of the excitation / emission



A fluorophore preferentially absorbs photons when the electric field vector is colinear to the absorption dipole. Emission is polarized along the emission dipole

Measuring fluorescence polarization reports on the rotation of the fluorophore (ex : lipids in different lipidic phase)



Fluorescence & Labeling Strategies

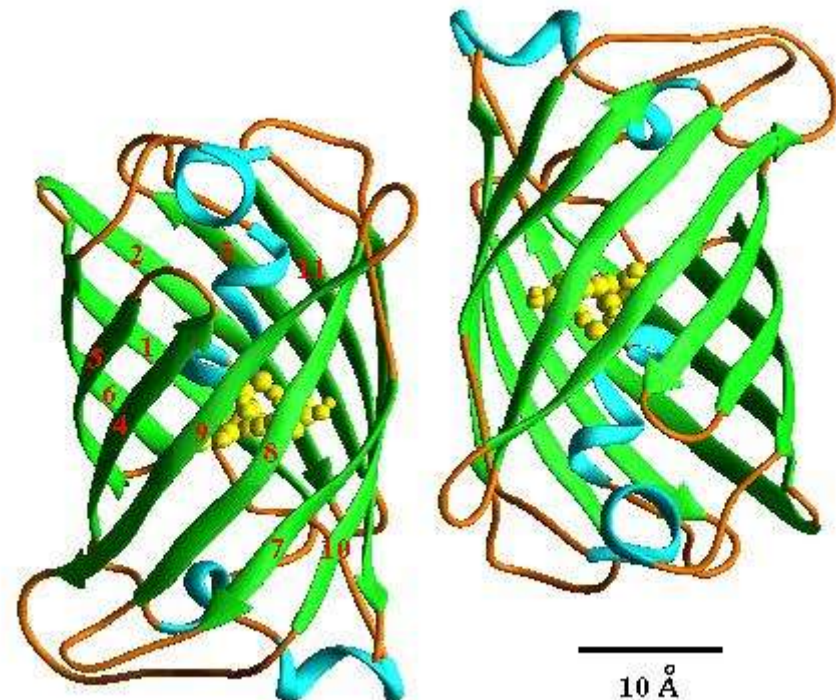
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Introduction to GFP

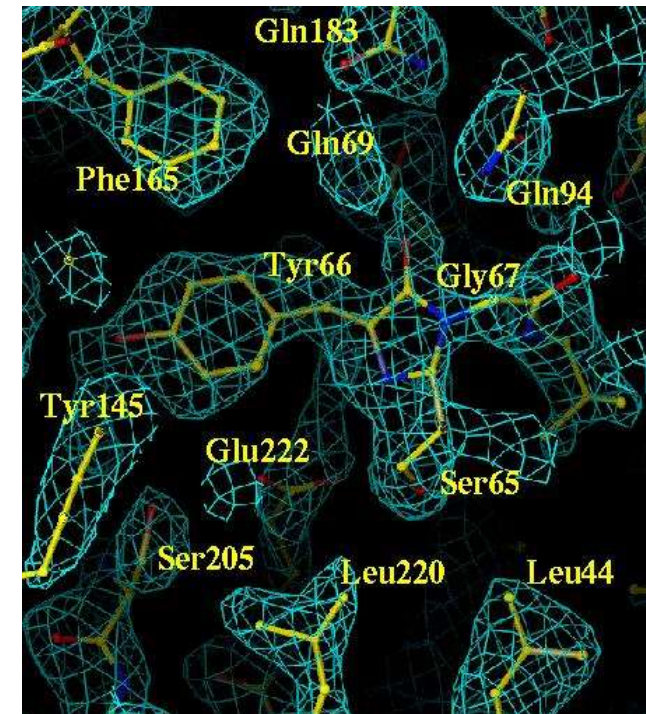
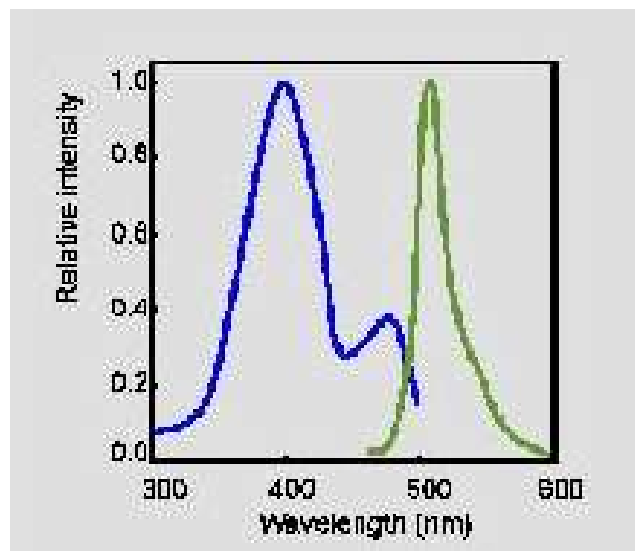
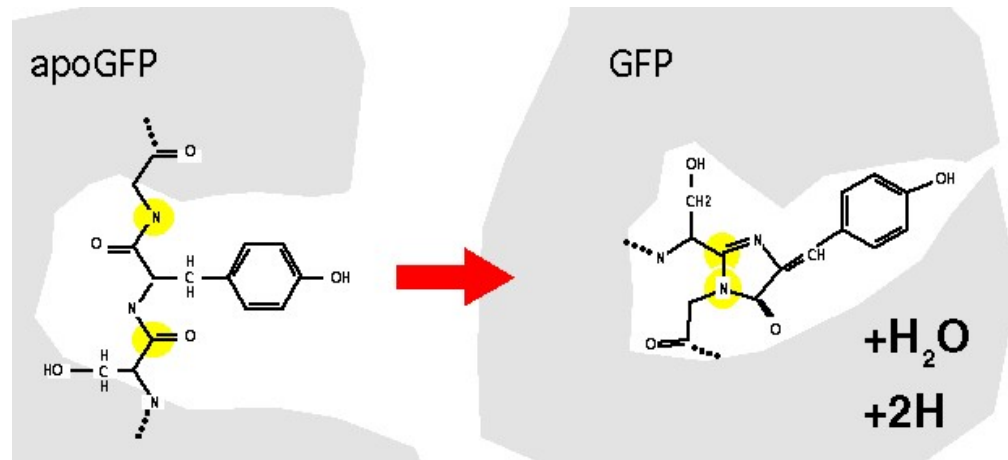
Green Fluorescent Protein :

- from *Aequorea victoria* jellyfish (1962, Shimomura et coll)
- gene cloned in 1992 by Prasher et coll.
- 238 AA protein, compact and globular. MW : 27 kDa
- monomeric at low concentrations



Introduction to GFP

Ser65, Tyr66 and Gly67 → maturation



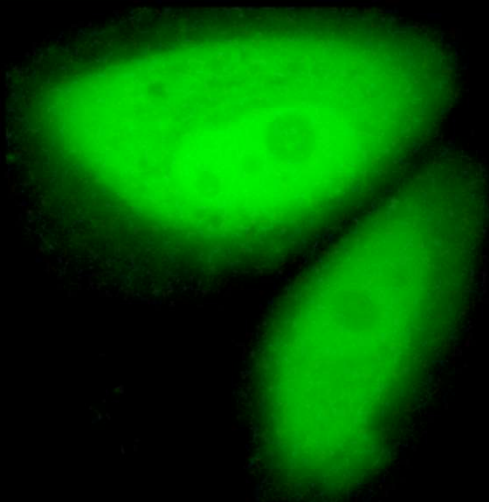
Protection of the fluorophore within the barrel allows for a relatively good photostability



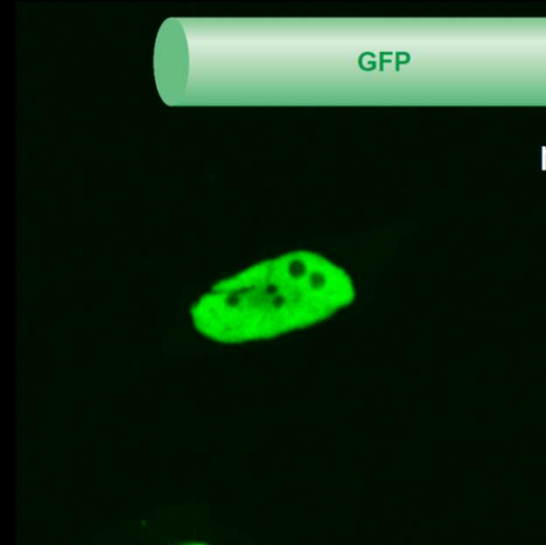
Examples of FP targeting to subcellular organelles



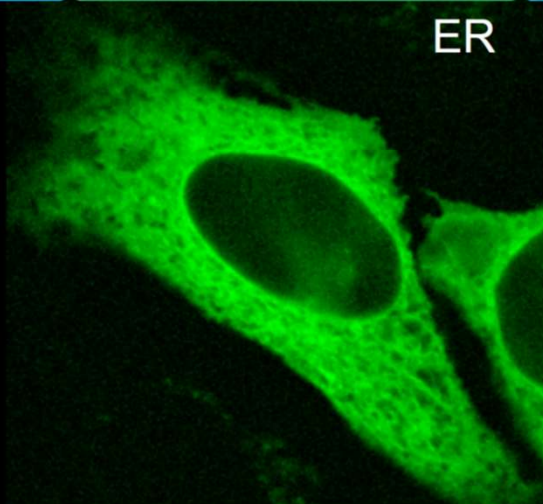
(free)



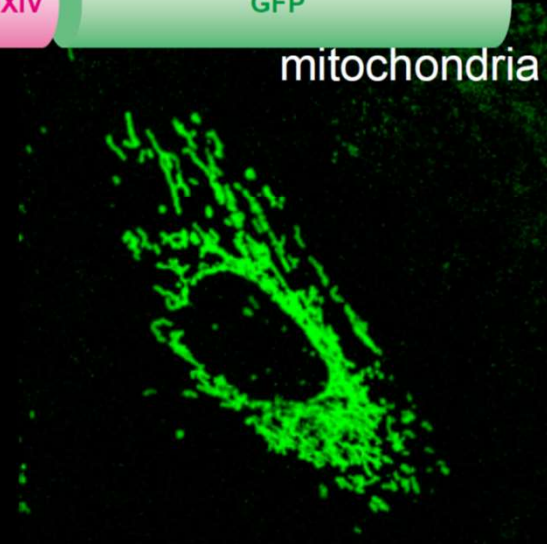
nucleus



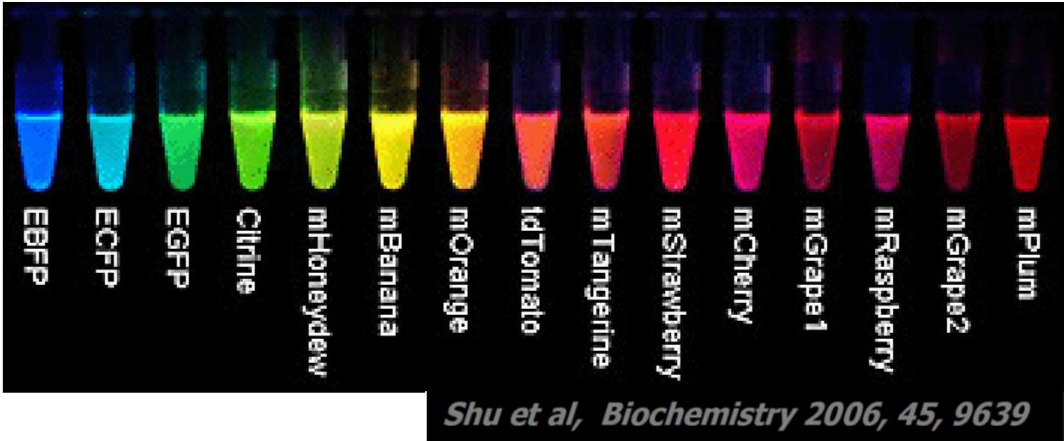
ER



mitochondria



Creating new FPs



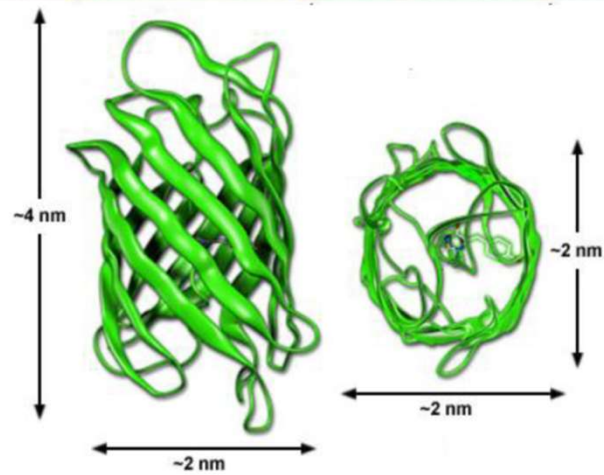
MANY VARIANTS OF THE GFP HAVE BEEN DEVELOPPED:

- higher brightness
- higher photostability
- less pH sensitive
- better eukaryotic / procaryotic expression
- monomeric forms
- photoactivable / photoconvertible forms



The two families of FPs

Hydrozoan FPs



Anthozoan FPs

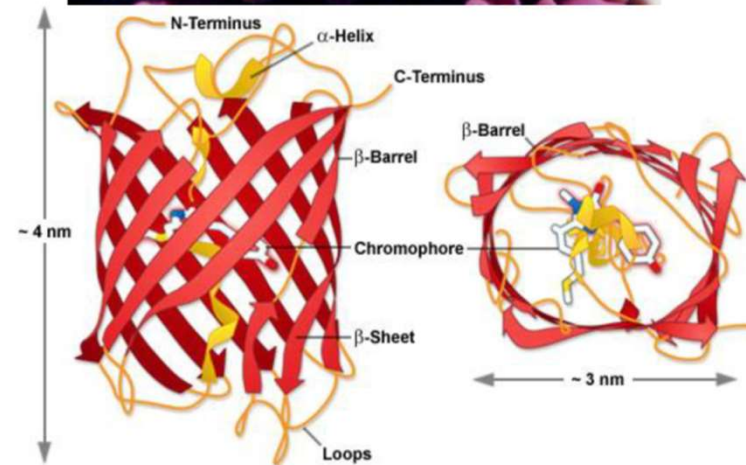


Figure 1 displays a scatter plot of protein brightness (Y-axis, 0 to 130) versus emission wavelength (X-axis, 400 to 750 nm). The plot shows a large number of data points, each representing a protein. The points are color-coded based on their emission wavelength, ranging from blue (shorter wavelengths) to red (longer wavelengths). The plot includes a legend for 'Oligomerization' with a dropdown menu set to 'All'. To the right of the plot, there are interactive controls for various parameters: Excitation Wavelength (nm) (350 to 800), Emission Wavelength (nm) (350 to 800), Extinction Coefficient (10000 to 230000), Quantum Yield (0 to 1), and Brightness (0 to 125). Below these controls, there are two columns of buttons labeled 'Y Axis' and 'X Axis', each containing a list of parameters: Excitation λ, Emission λ, Stokes Shift, EC, QY, Brightness, pKa, Stability, Maturation, and Lifetime. The 'Brightness' button is highlighted in the 'Y Axis' column.



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Fluorescence & Labeling Strategies

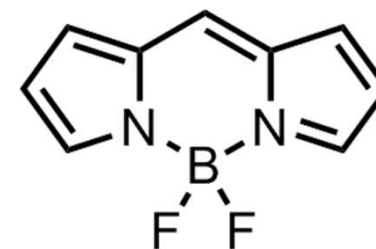
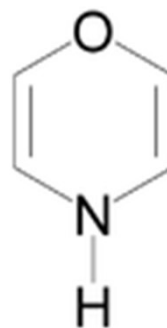
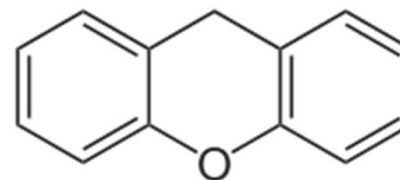
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Organic Fluorophores

- Various chemical structures

- Xanthenes
- Cyanines
- Oxazine
- Bodipy
- Perylene
- Naphthalene
- Coumarine
- Acridine
-



Organic fluorophores

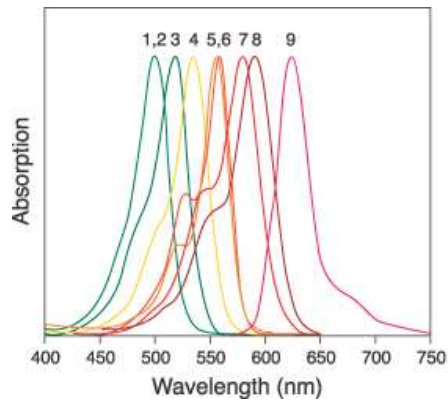
- Various brand names
 - Cy dyes (Cy 3, Cy5) : Amersham
 - Alexa fluor (Invitrogen)
 - DyLight (Thermo / Pierce)
 - Atto dyes (Atto-Tec)
 - Fluoprobes (Interchim)
 - Abberior
 - Dy and megastockes (Dyomics)
 -

**THERE IS NO CORRELATION BETWEEN THE NAME
AND THE STRUCTURE, AND THUS, THE
PROPERTIES !!!**

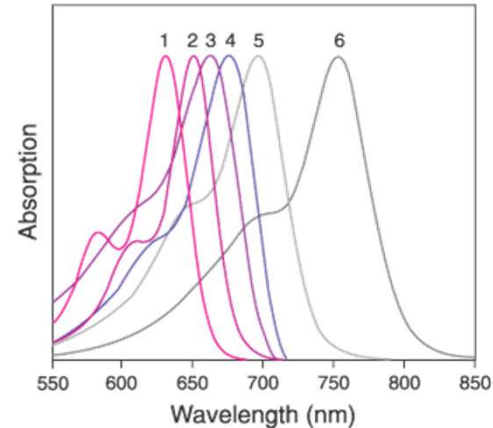


Example of Alexa dyes

Derivatization of existing dyes to make them more soluble, more photostable, or to circumvent patents...

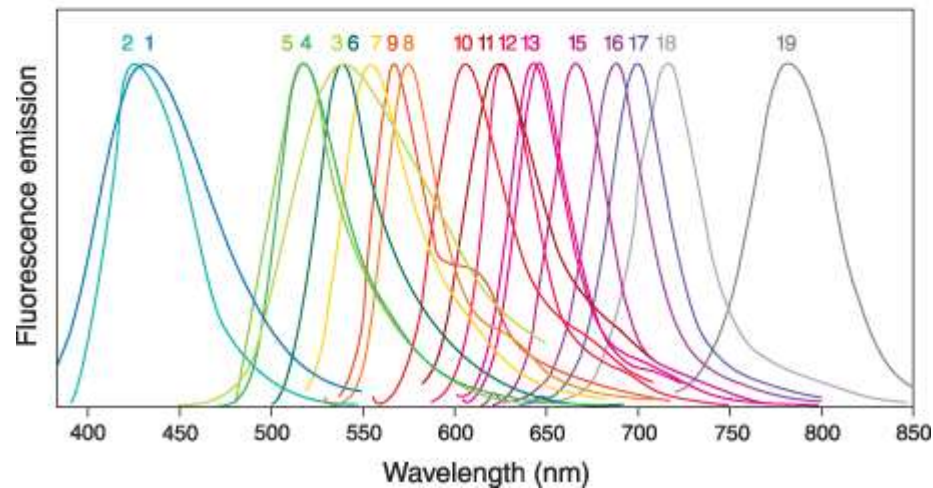


- 1 Alexa Fluor 488
- 2 Alexa Fluor 500
- 3 Alexa Fluor 514
- 4 Alexa Fluor 532
- 5 Alexa Fluor 555
- 6 Alexa Fluor 546
- 7 Alexa Fluor 568
- 8 Alexa Fluor 594
- 9 Alexa Fluor 610



- 1 Alexa Fluor 633
- 2 Alexa Fluor 647
- 3 Alexa Fluor 660
- 4 Alexa Fluor 680
- 5 Alexa Fluor 700
- 6 Alexa Fluor 750

- | | | | |
|--------------------|---------------------|---------------------|---------------------|
| 1. Alexa Fluor 350 | 6. Alexa Fluor 514 | 11. Alexa Fluor 594 | 16. Alexa Fluor 660 |
| 2. Alexa Fluor 405 | 7. Alexa Fluor 532 | 12. Alexa Fluor 610 | 17. Alexa Fluor 680 |
| 3. Alexa Fluor 430 | 8. Alexa Fluor 546 | 13. Alexa Fluor 633 | 18. Alexa Fluor 700 |
| 4. Alexa Fluor 488 | 9. Alexa Fluor 555 | 14. Alexa Fluor 635 | 19. Alexa Fluor 750 |
| 5. Alexa Fluor 500 | 10. Alexa Fluor 568 | 15. Alexa Fluor 647 | |



How to turn a dye into a useful tool ?

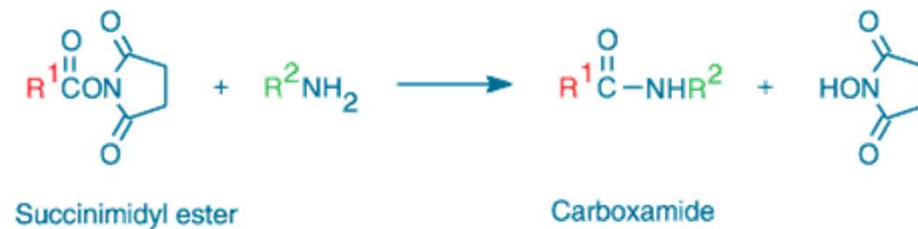
- To be useful, your fluorophores needs to have some « activity »
 - DNA Intercalating dyes
 - Dyes attached to specific small molecules (ex : lipids)
 - Dyes need to be congugated to your protein of interest (for in vitro experiments), to an antibody or a nucleic acid (for your cell microscopy experiments)

How to label your molecule of interest with fluorophores

REACT ON PRIMARY AMINES

Proteines : N-terminal or lysine

Oligonucleotides : NH_2 added during sythesis



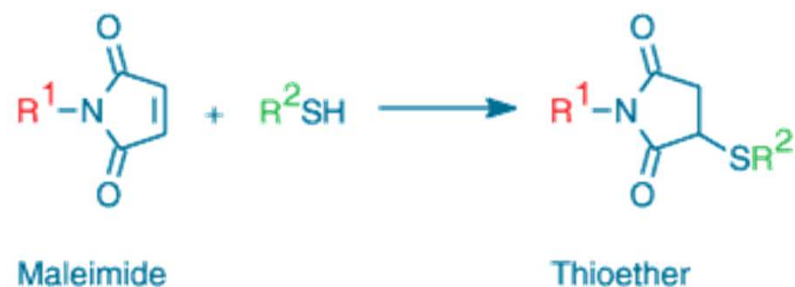
Around pH7, the lyzines are protonated, and thus only the N-terminus is reactive.



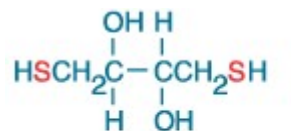
How to label your molecule of interest with fluorophores

REACT ON THIOLS

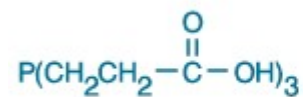
Proteins : cysteins



TO REDUCE CYSTEINES, USE TCEP



(A) dithiothreitol (DTT)



(B) tris-(2-carboxyethyl)phosphine, hydrochloride (TCEP)



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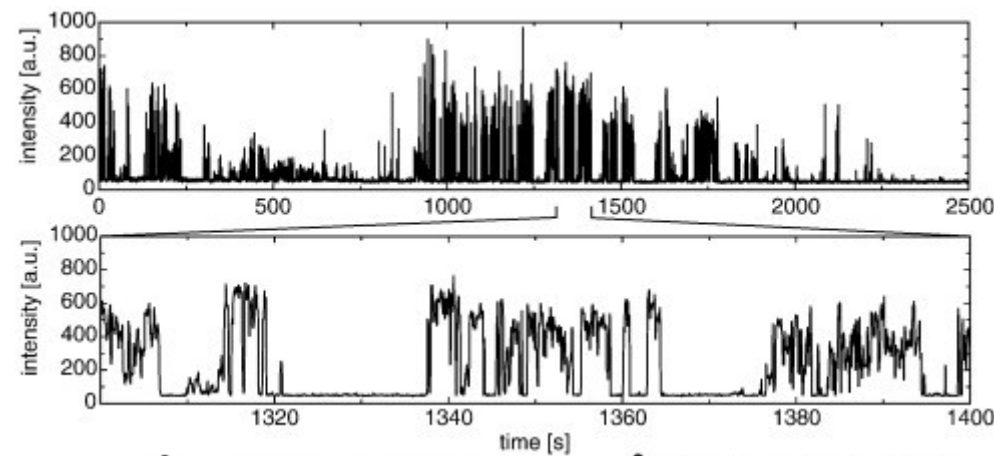
Photodestruction of the fluorophore :

- Mainly due to O_2
 - Interaction with long lived excited states (Triplet)
 - Production of radicals



Blinking

All fluorophores show blinking, except nitrogen vacancies in nanodiamonds, and some specially engineered QD

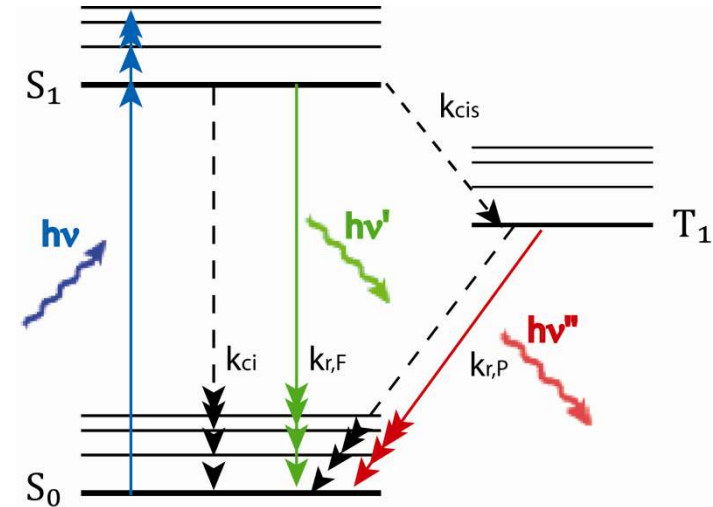


QD blinking at various time scales



Blinking

- Triplet blinking



- Random event
- Triplet state is long lived (μs - ms) : appear as blinking
- Quenched by O_2
- Quenched by reducing agents (MEA, BME, Trolox)

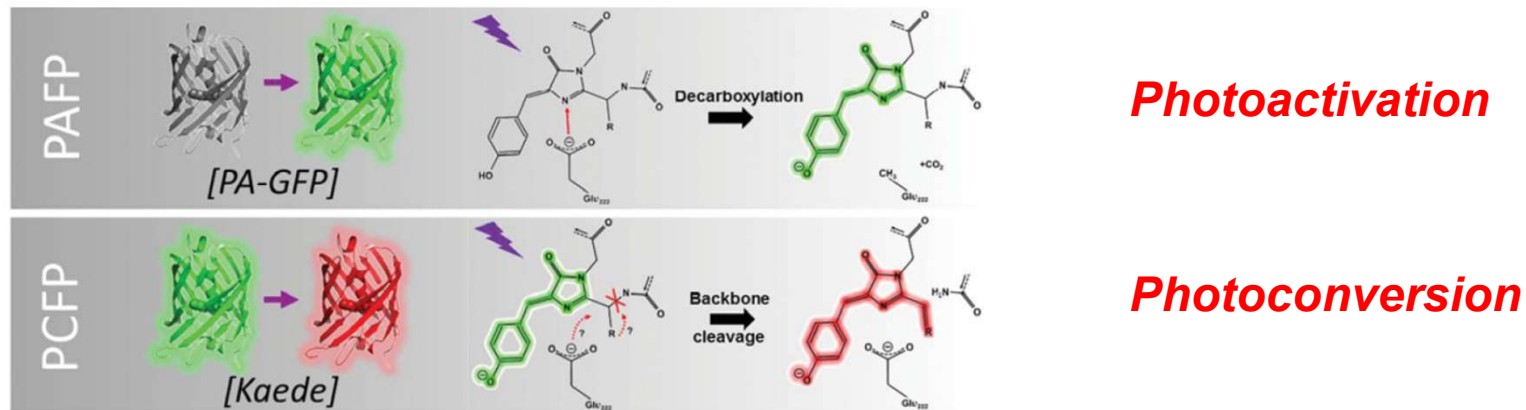


- Redox blinking
 - Photoinduced electron transfer
 - Higher probability in the triplet state
 - The ions are not fluorescent
 - Triplet state quenchers help restoring the ground state

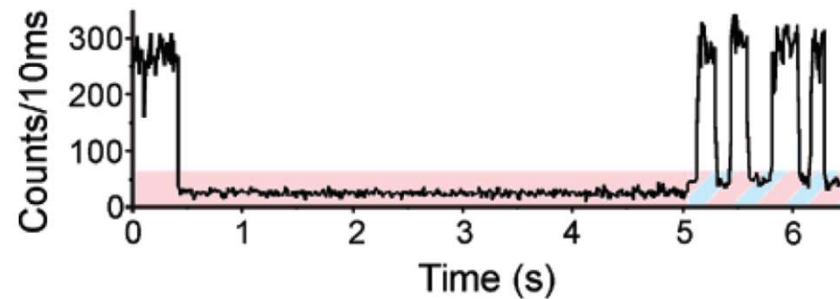
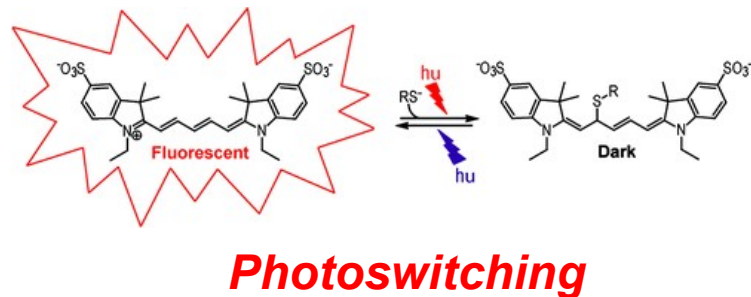


Photochromism

- photoinduced reaction between 2 chemical species
- Seen in the FPs

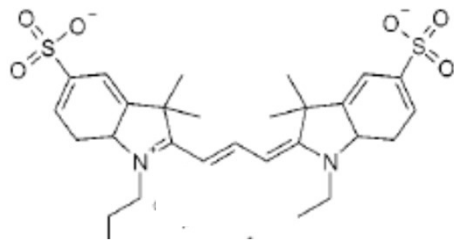


- Cy5 / thiol : dark state, that can be restored by UV light

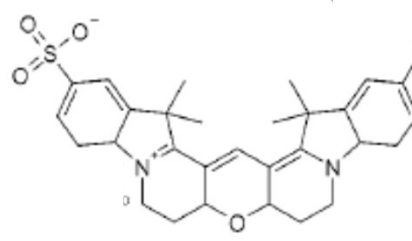


Photoisomerization

– Cyanines



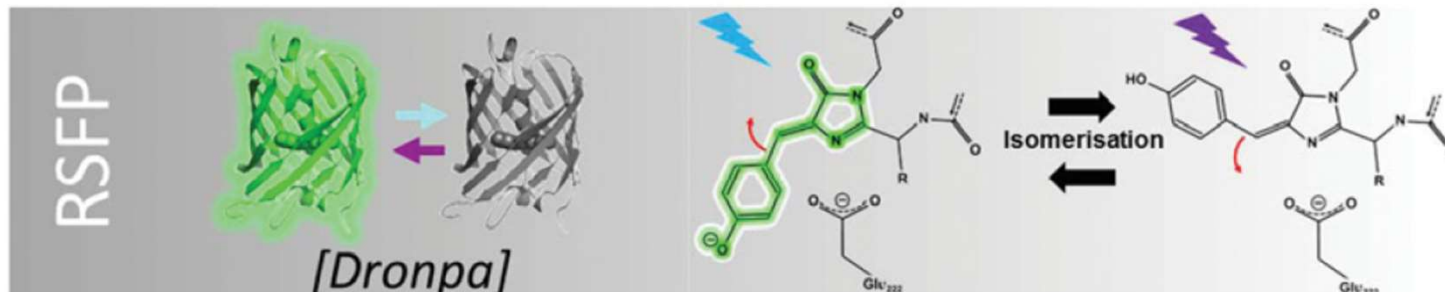
Cy3 : only the *cis* is fluorescent



cy3B

- Rigidified version
- no cis-trans isomerization
- higher QY

– FPs

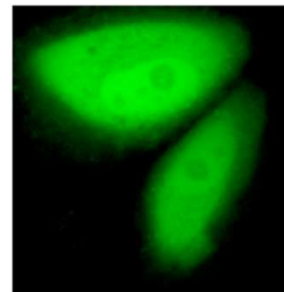
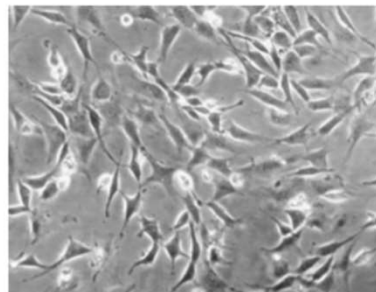
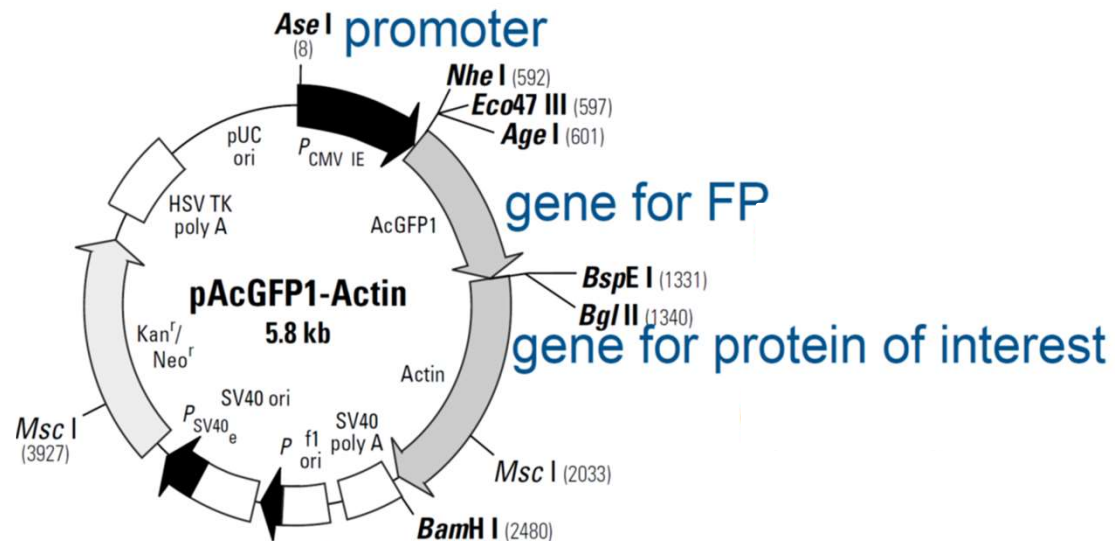


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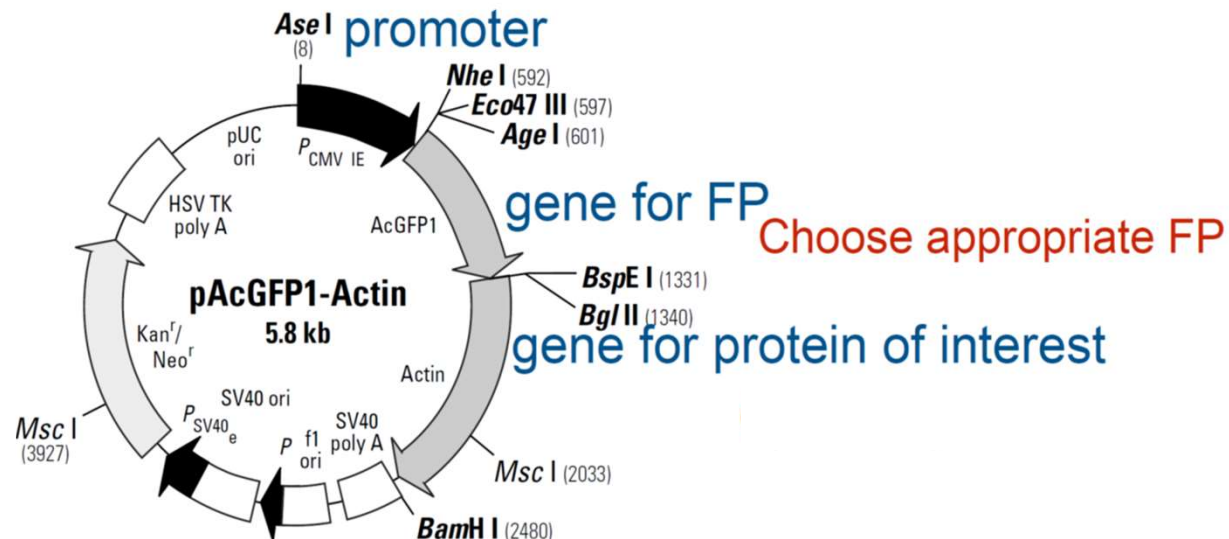
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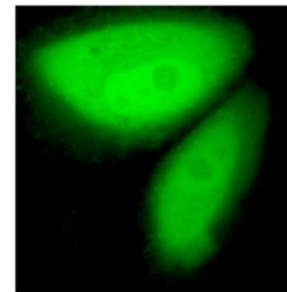
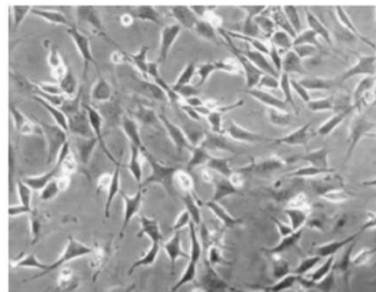
Labeling your POI with a GFP



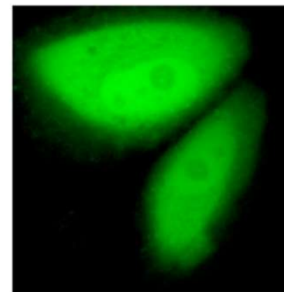
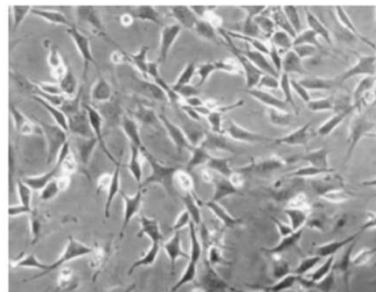
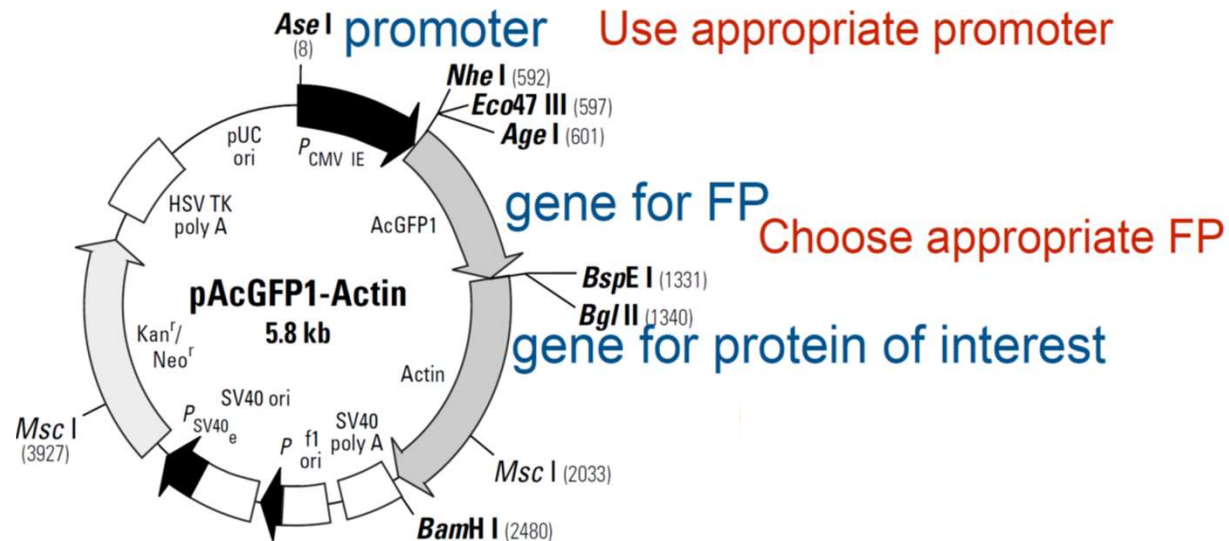
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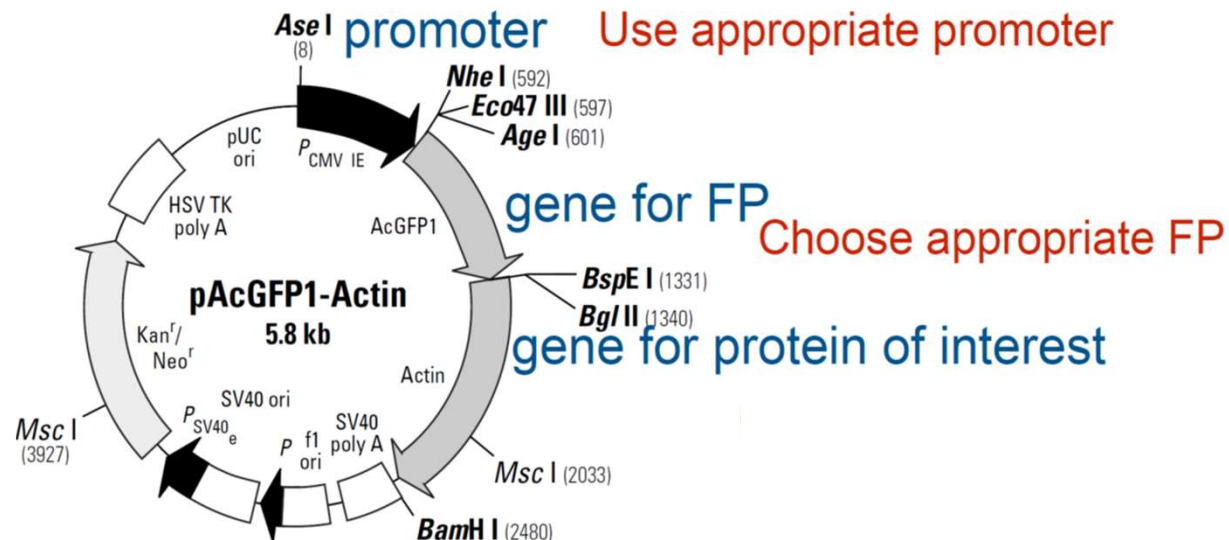
Choose appropriate FP



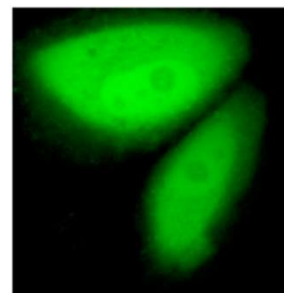
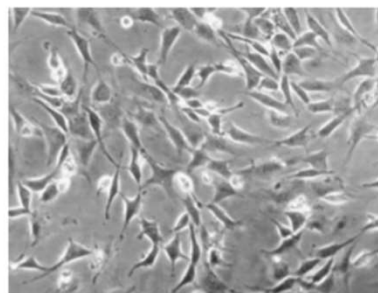
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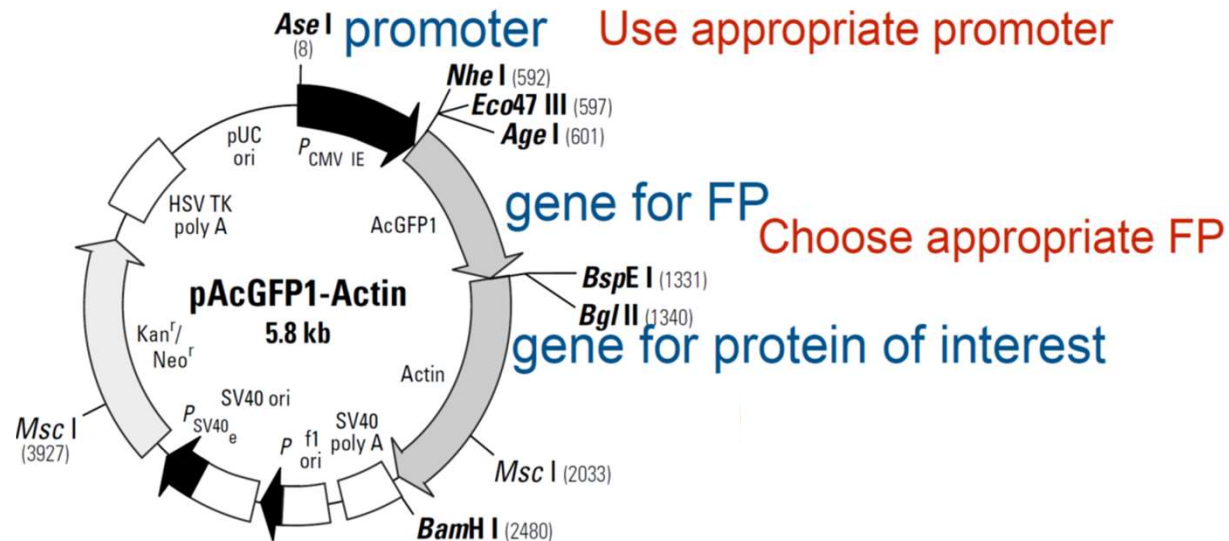
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How to introduce?



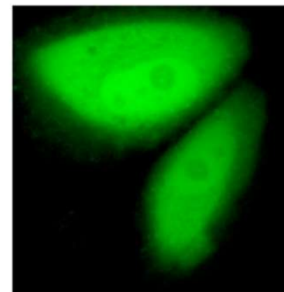
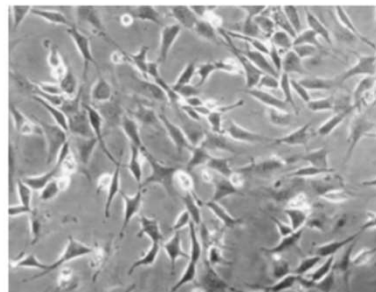
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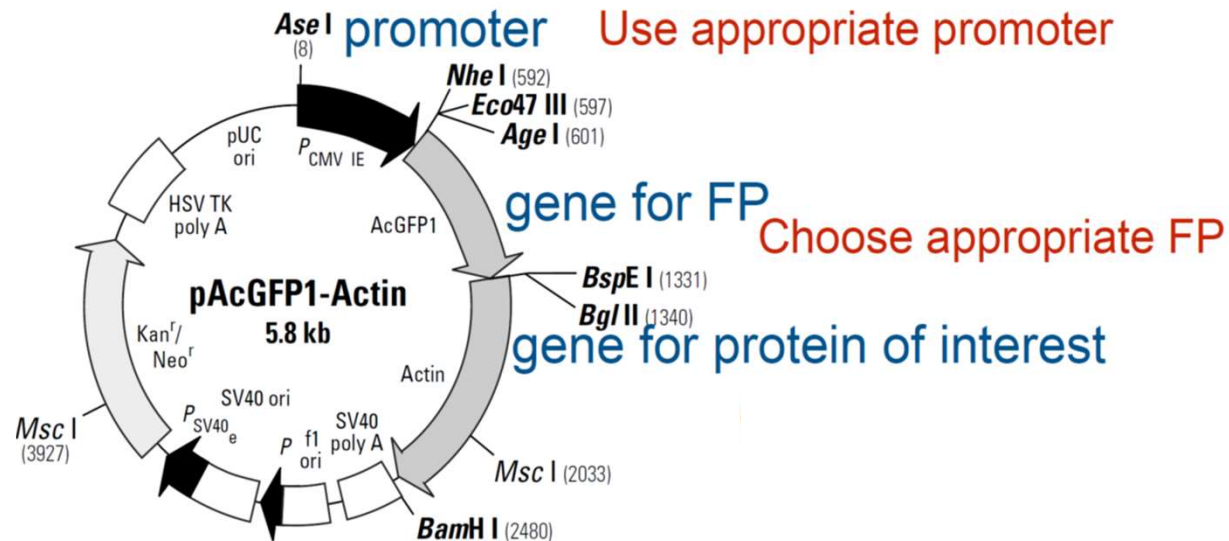
How to introduce?



Expression level is crucial



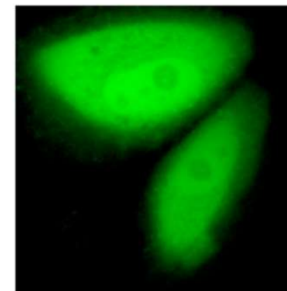
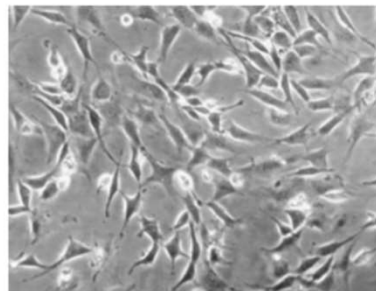
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How to introduce?

Expression level is crucial

Does labelled protein conserve its function?

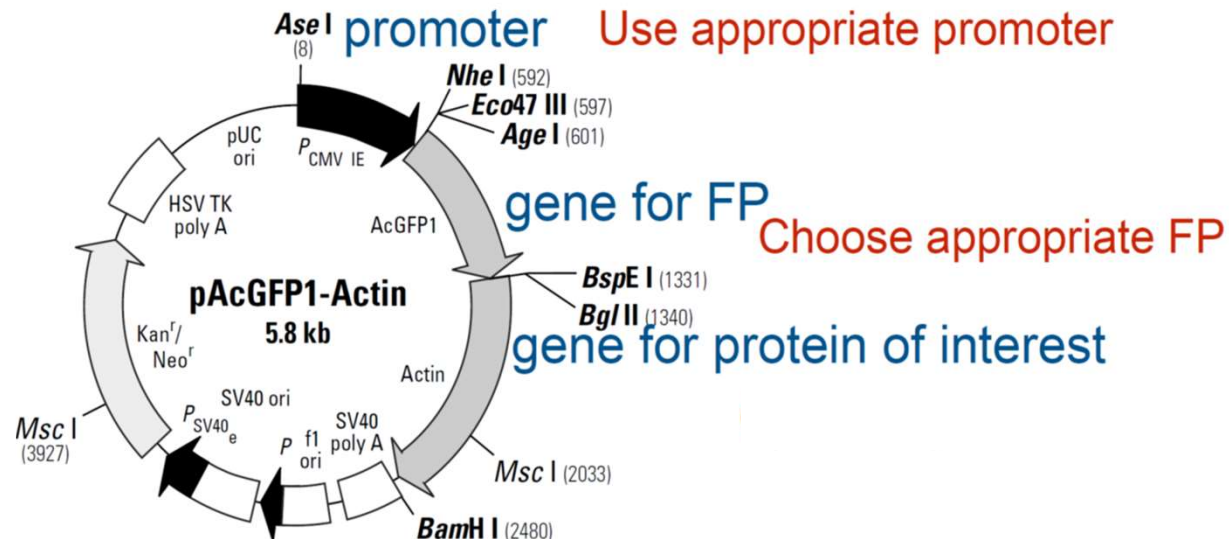


No competition to endogenous protein?

No impact to cellular function?



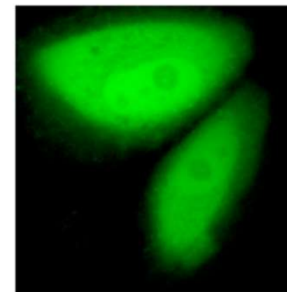
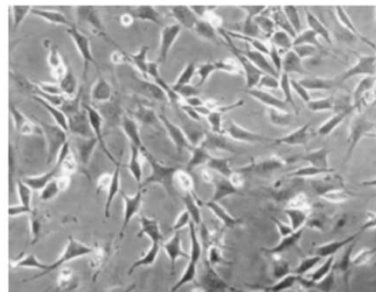
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How to introduce?

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Does labelled protein conserve its function?



No competition to endogenous protein?

No impact to cellular function?

These questions are valid for FPs, and for other labeling strategies



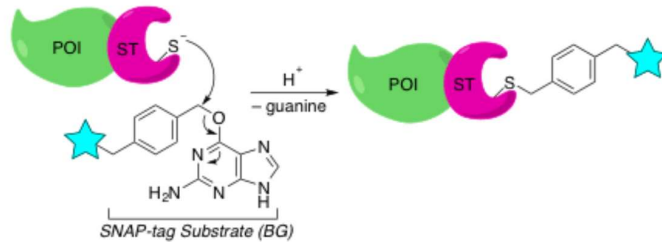
Self labeling Tags

Halo-Tag



- Express your POI as a fusion with the self labeling protein

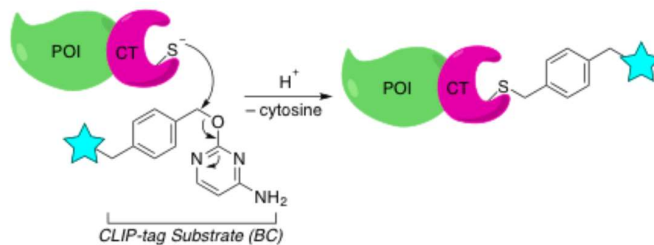
SNAP-Tag



- Incubate with the fluorophore having the right reactive group

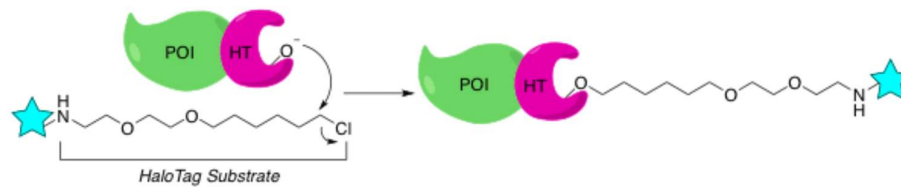
- Wash to remove excess dye

CLIP-Tag

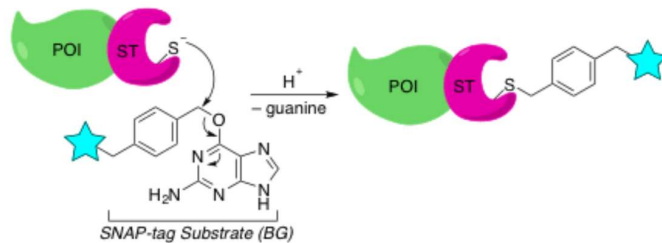


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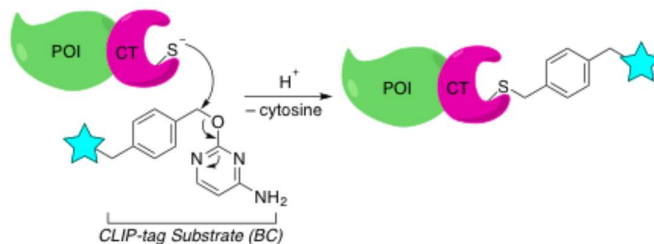
Halo-Tag



SNAP-Tag



CLIP-Tag



- Broad choice of fluorescent dyes
- Higher fluorescence intensity and photostability
- Fluorescence initiated when the label is added (temporal control)
- A single construct is sufficient to label with different fluorophores (pulse chase)
- Live cell labeling

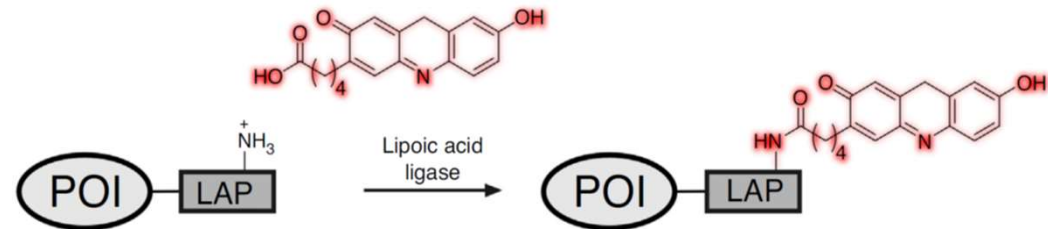
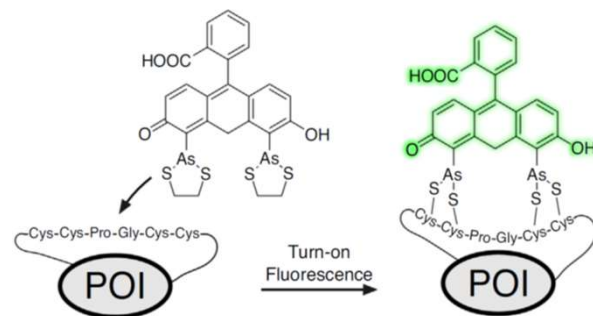


- Non-specific staining
- Bulky (20-33 kDa) fusion
- N or C-ter fusion preferred



Peptide-based labeling Tags

His tag – Ni ²⁺ – NTA	One-step	Non-covalent	10 amino acids
D4 tag – DpaTyr	One-step	Non-covalent or covalent	4–12 amino acids
Tetracysteine (CCXXCC)	One-step	Covalent	6 amino acids
dC10 α	One-step	Covalent	22 amino acids
Sortase A	Two-step	Covalent	6 amino acids
LplA	One-step or two-step	Covalent	13 amino acids



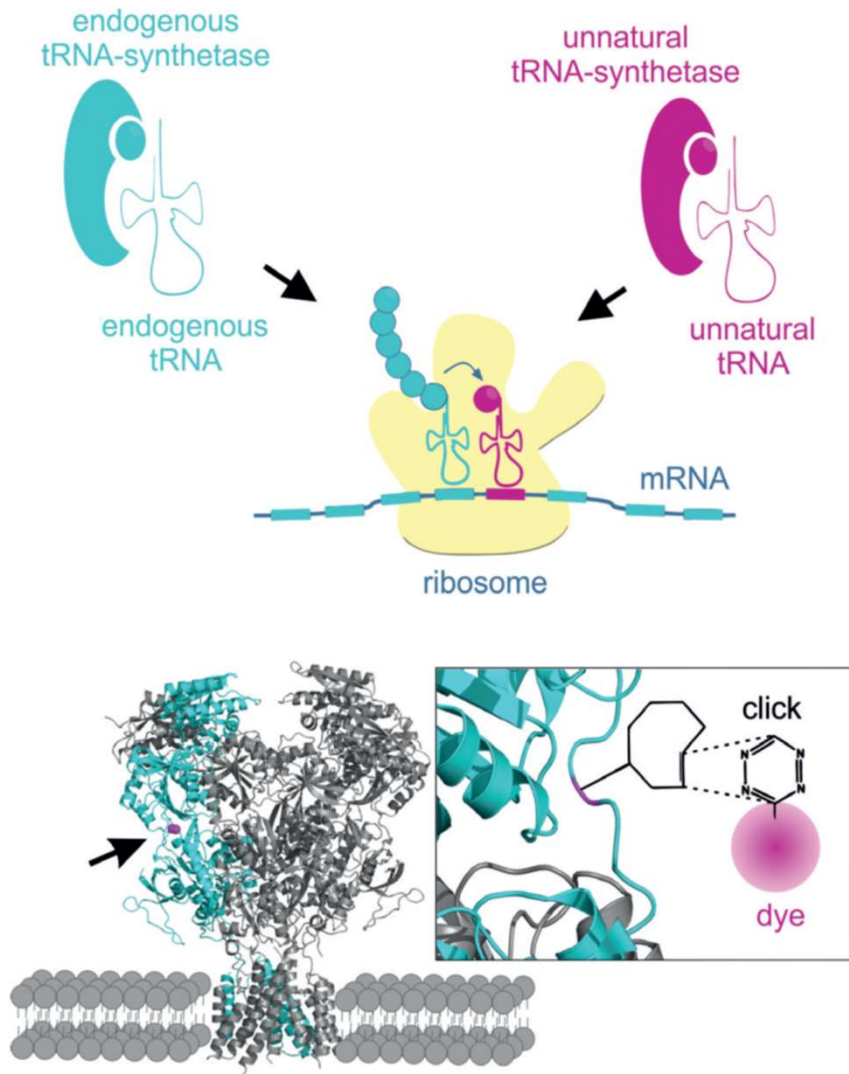
- Small size of the fusion
- Fluorescence initiated when the label is added (temporal control)
- Live cell labeling



- Very limited choice of dyes
- Non-specific staining
- N or C-ter fusion required (except Tetracys & LplA)
- No experience at CBS ☺



Labeling unnatural aminoacids

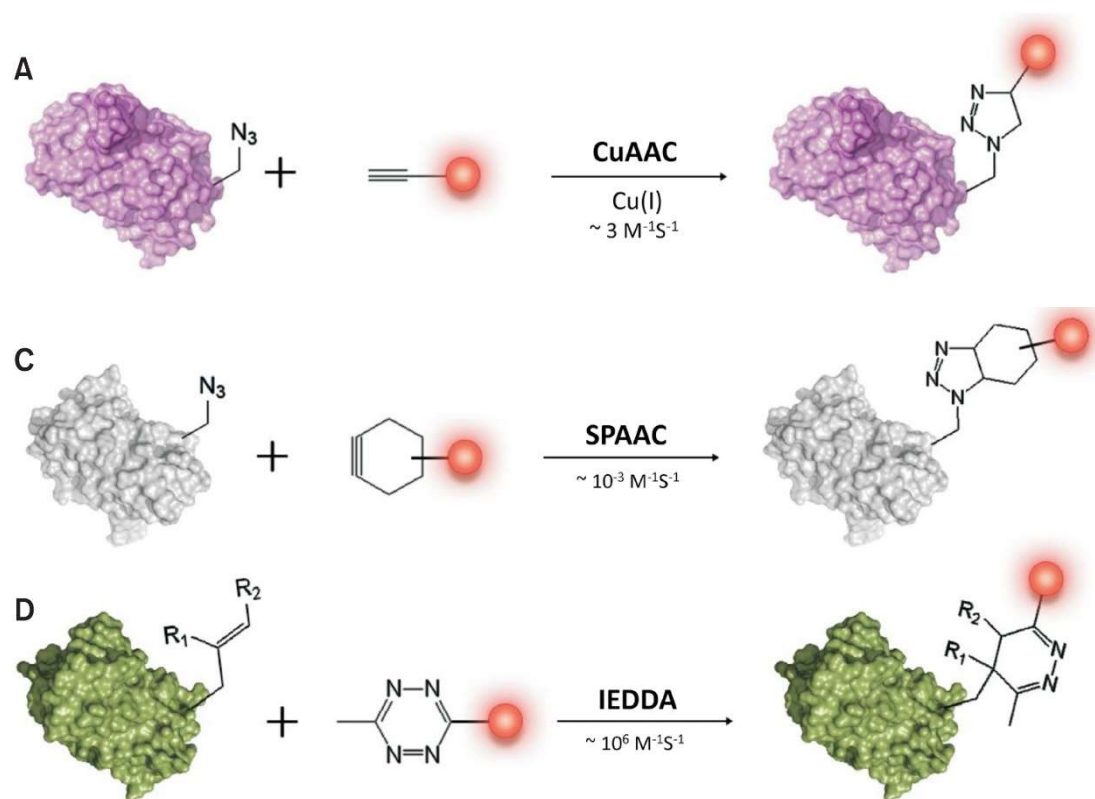


- Smallest size of the fusion
- Fluorescence initiated when the label is added (temporal control)
- Vast (and increasing) choice of fluorophores
- Orthogonal labeling possible
- Live cell labeling (beware of Cu^{2+})

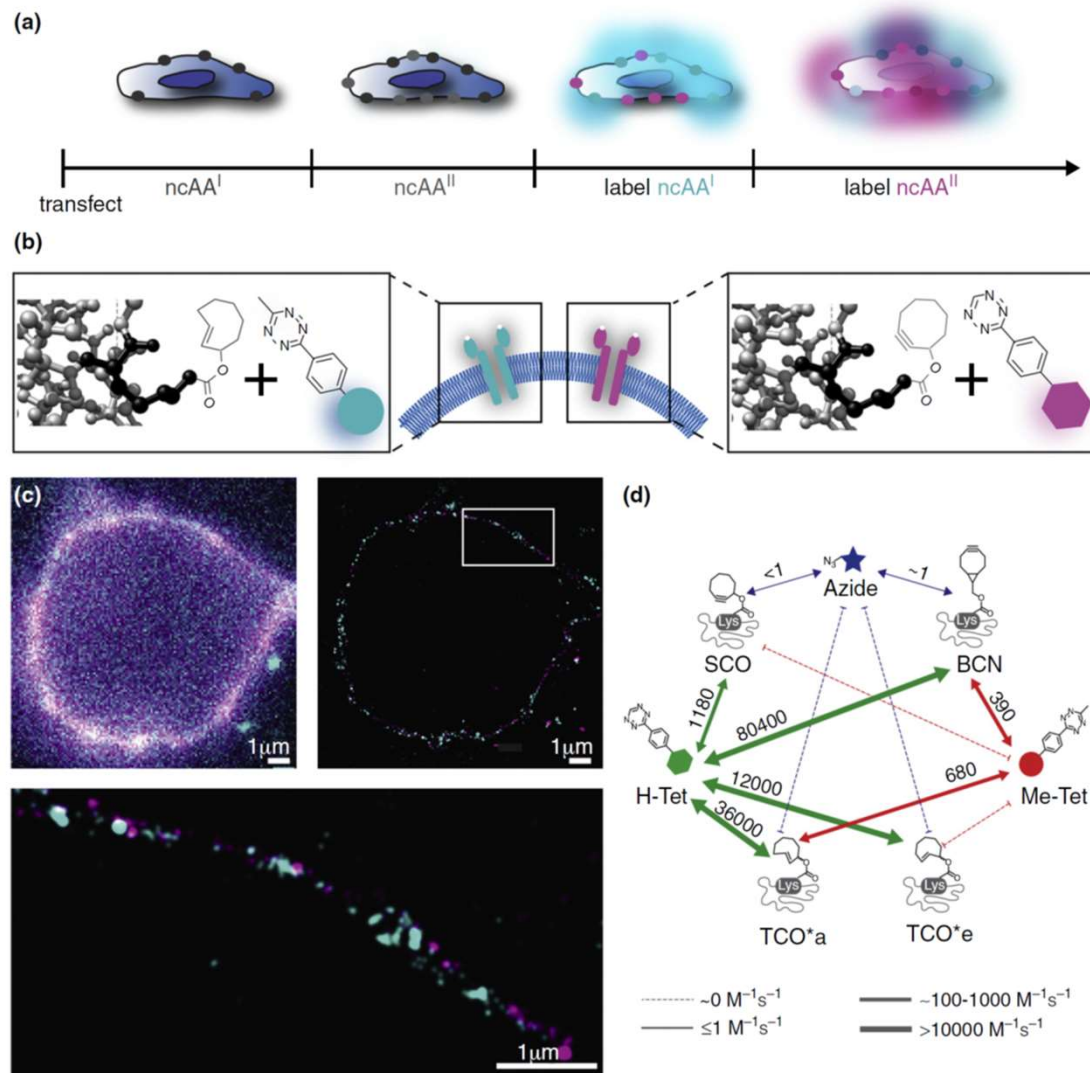
- Non-specific staining
- Non specific incorporation of the UAA
- Reagents are expensive



Labeling unnatural amino acids



Labeling unnatural amino acids

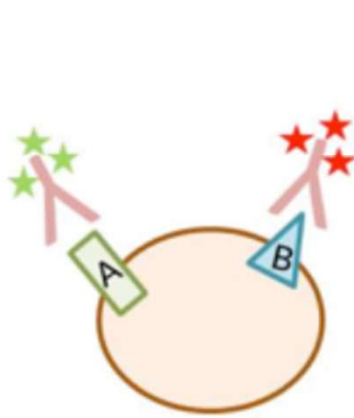


Fluorescence & Labeling Strategies

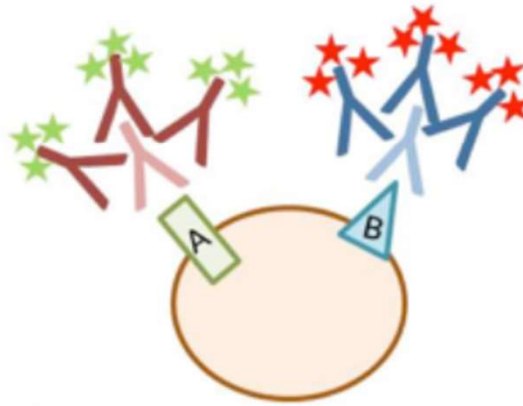
- *Basis of fluorescence & Properties of fluorophores*
- *Fluorescent proteins*
- *Organic dyes*
- *Fluorophores photophysics*
- *Protein labeling in live cells (FPs, Self-labeling tags, UAA)*
- ***Labeling in fixed cells (Immunostaining, non-specific stains (mb, nucleic acids, etc...), FISH***



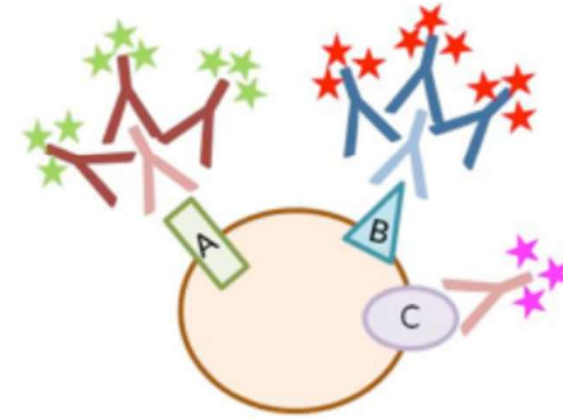
Immuno-fluorescence (IF)



Direct IF



Indirect IF



Combination of both



- Endogenous proteins
- Highly flexible
- Multi-colour labeling (if Abs are raised in different hosts)
- Signal amplification



- Fixed cells
- Antibody availability + epitope accessibility
- Non-specific background
- Size of antibody => loss in spatial resolution / image reconstruction

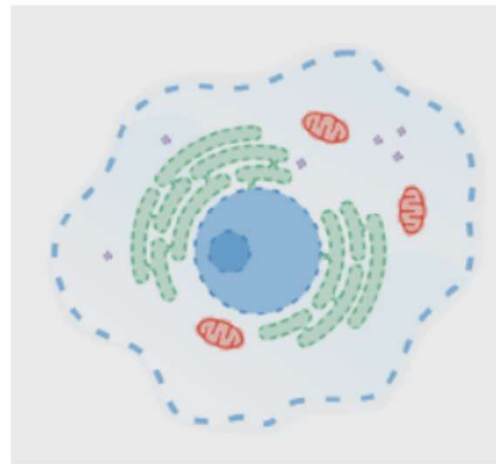


Immuno-fluorescence : experimental workflow



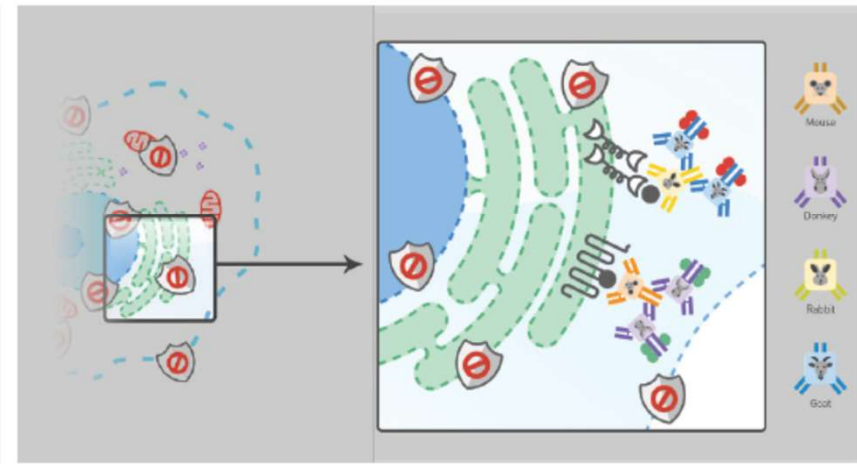
1) Chemical fixation

PFA, glutaraldehyde, methanol



2) Permeabilization

Detergents (Triton, SDS, saponin, digitonin...)



3) Blocking

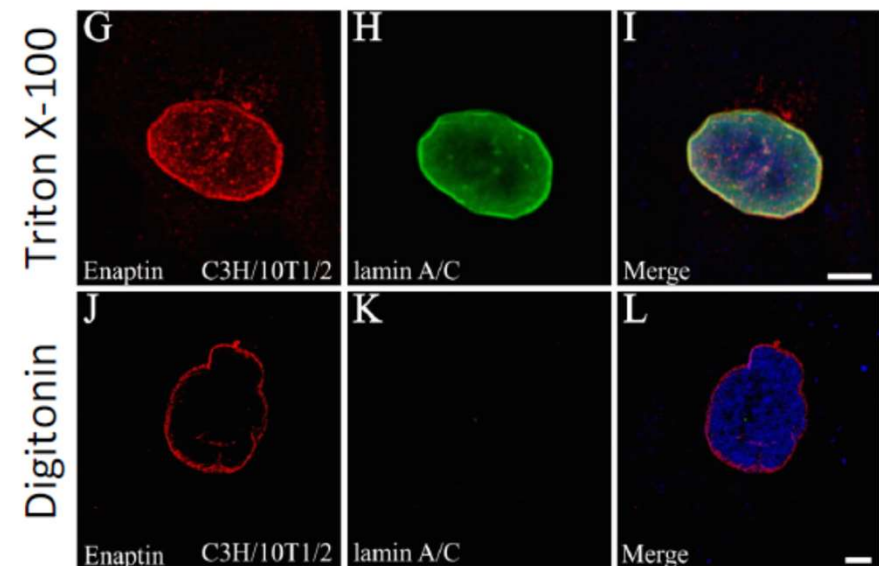
BSA, goat serum...

4) Incubation with primary $\pm$ secondary antibody

Variables to optimize: fixation agent, time and temperature, detergents, blocking agent, antibody concentration and incubation time

These depend on

- sample
- antibody
- imaging method

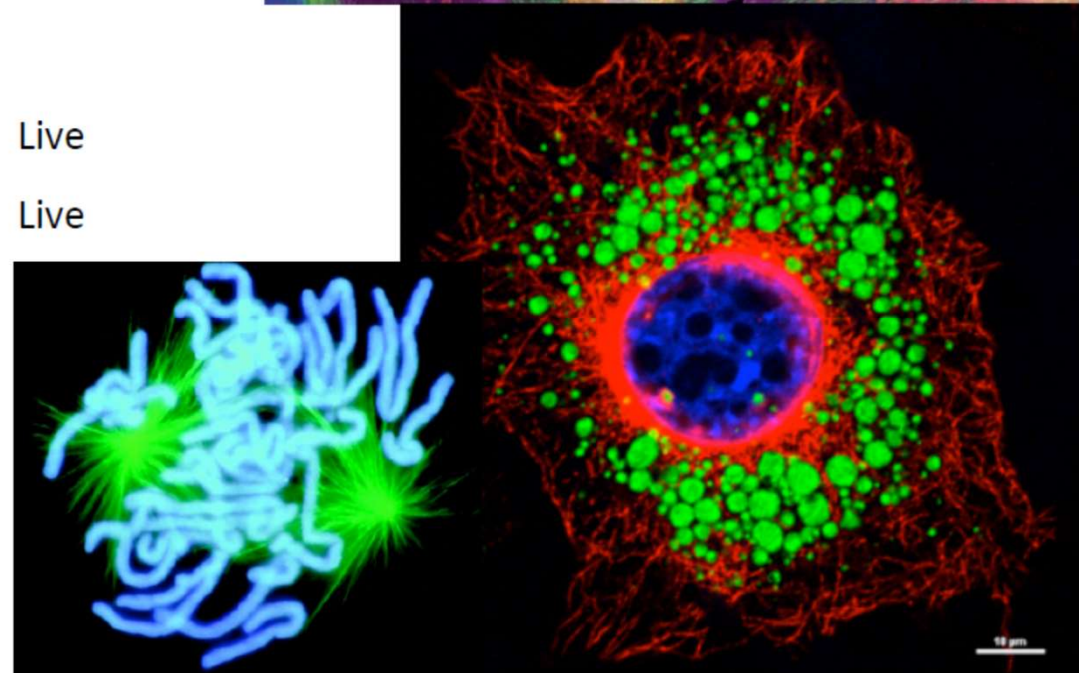
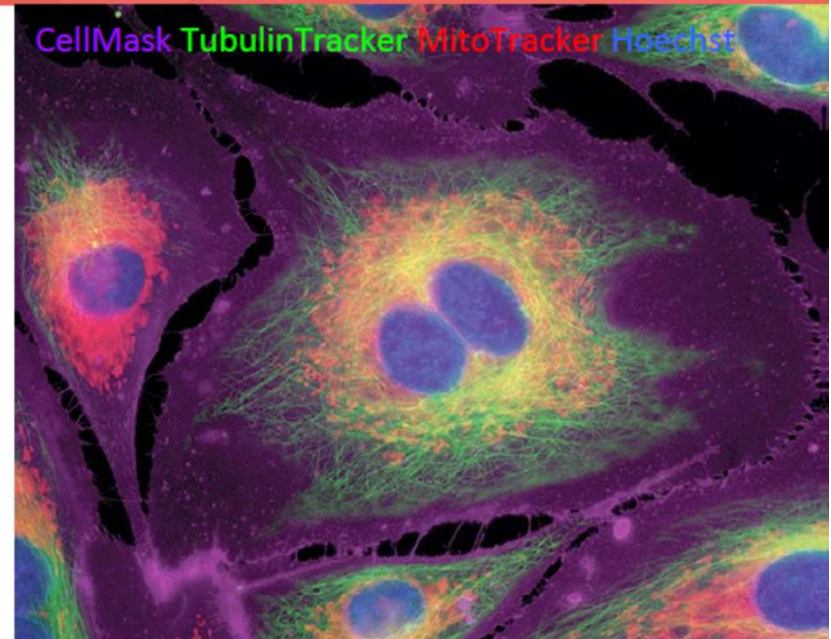


Chang *et al*, 2015

Organelle-specific labelling

Ready-to-use reagents

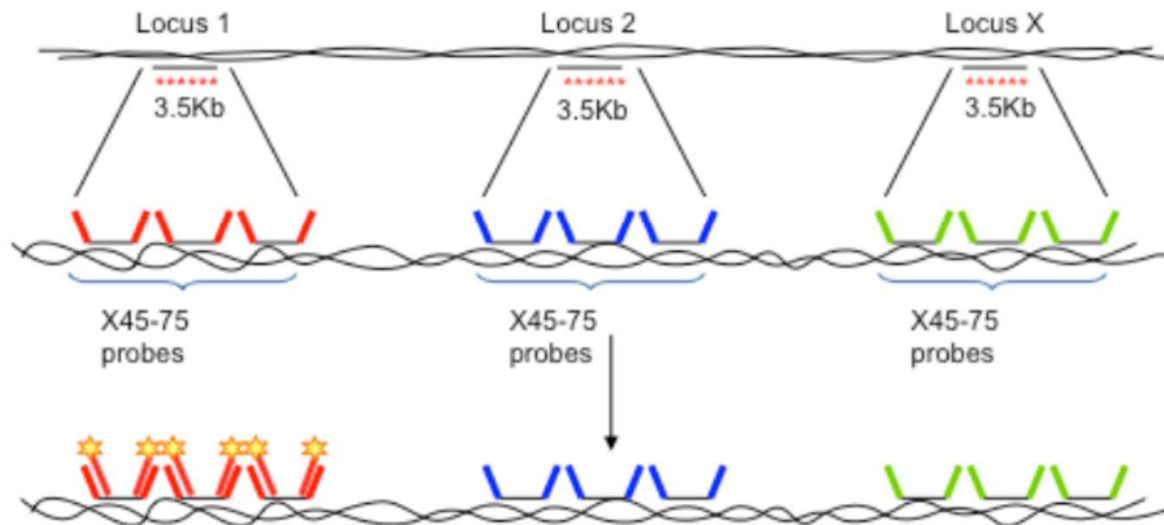
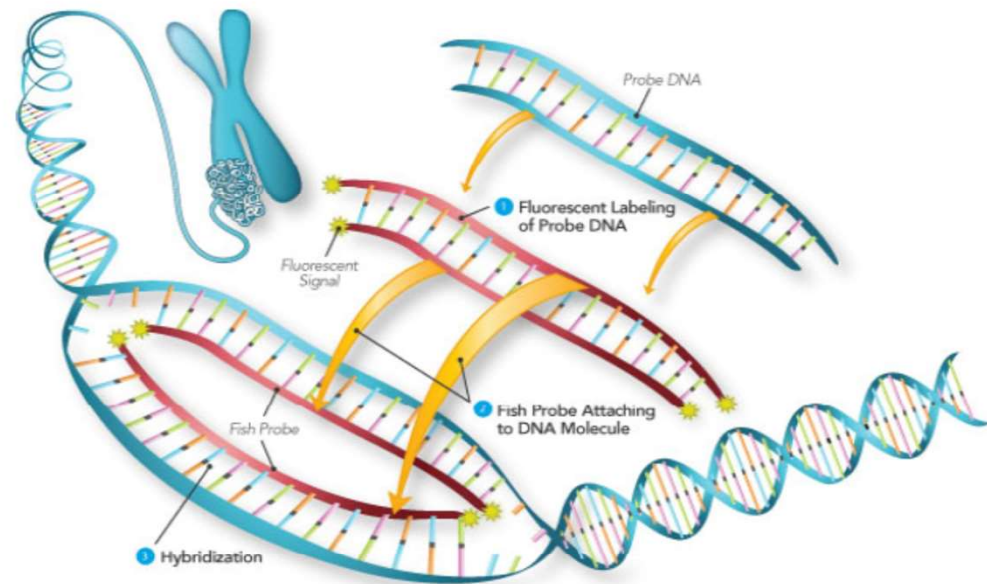
Organelles	Reagents	
Chromatin	DAPI, Hoechst, SYTO, DRAQ5	Live, fixed
Actin filaments	phalloidin	Fixed
Plasma membrane	FM64, CellMask	Live, fixed
Lipid bilayers	DiOC, DiIC, BODIPY-modified lipids...	Live, fixed
ER	ER-Tracker	Live
Mitochondria	Mito-Tracker	Live
Golgi	BODIPY Cer	
Lipid droplets	Nile Red, LipidTOX	
Ion indicators	BAPTA AM, Fluo3-AM	



Labeling specific genomic loci: DNA FISH

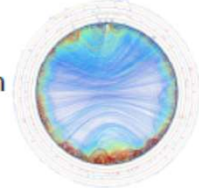
Fluorescent *in situ* hybridization

- 1- Design a set of DNA probes for target locus
- 2- Fix cells
- 3- Denaturation and annealing
- 4- Post-fixation
- 5- Imaging

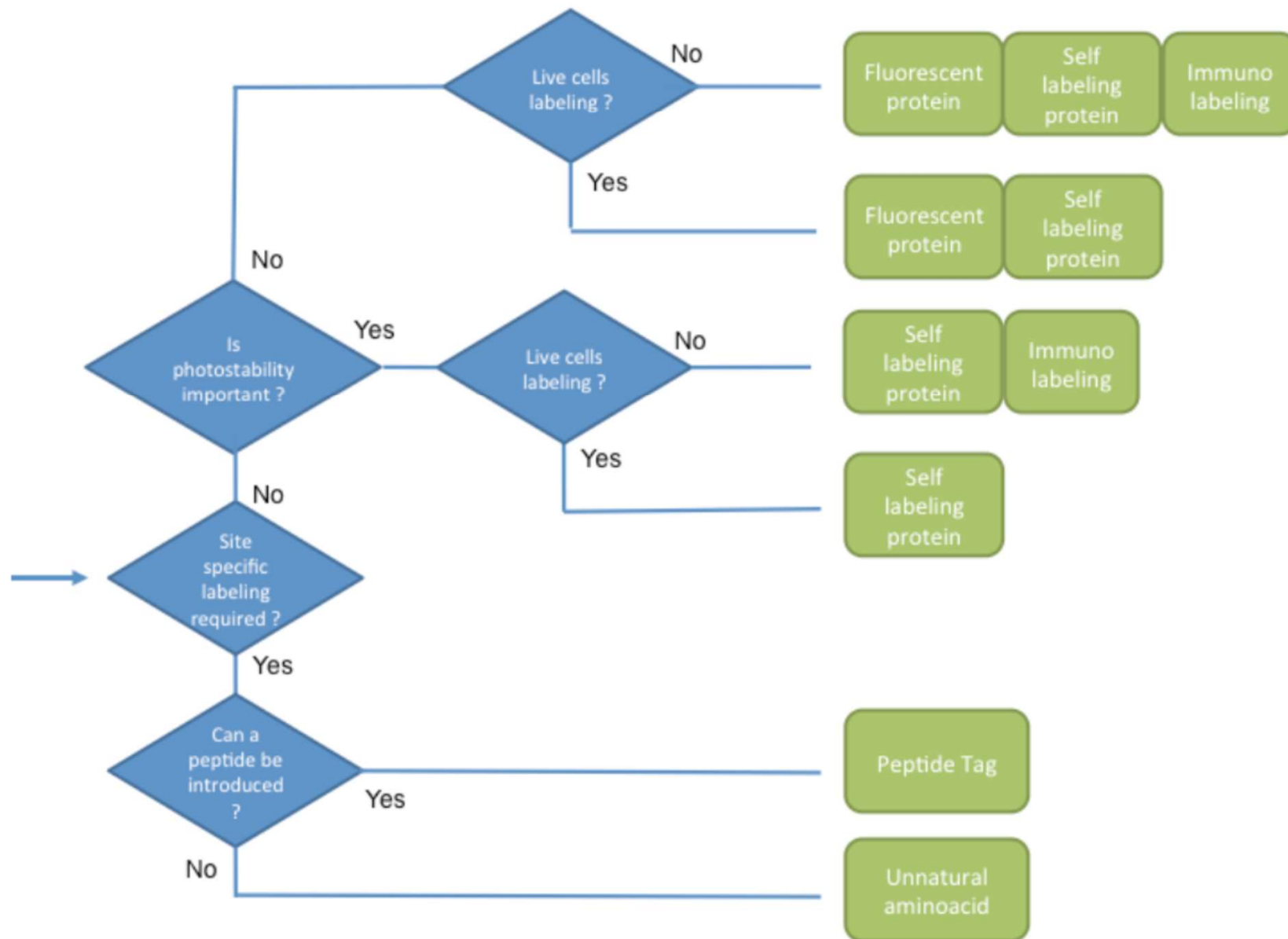


Multiplexed DNA FISH

Marcelo Nollmann's team

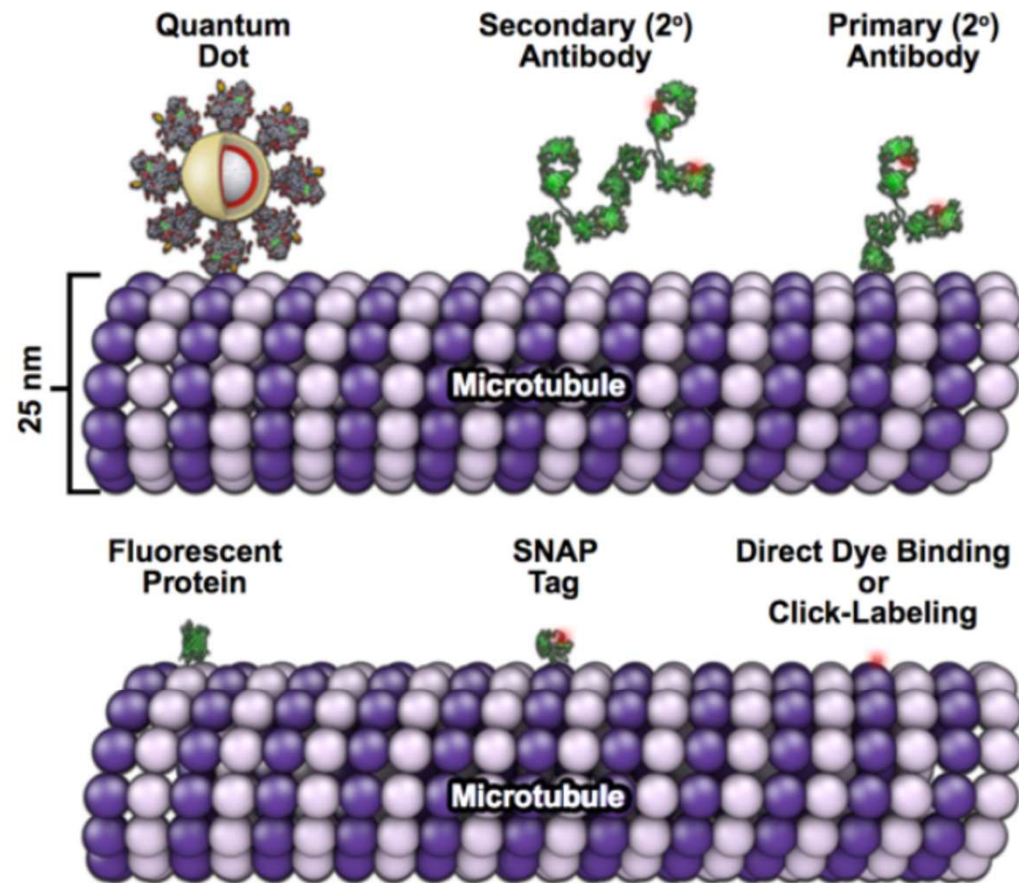


What should I ask myself before choosing a labeling strategy ?



If you are still hesitating...

- Size of the label
- Displacement of the probe from the target



→ This can be critical for super resolution microscopy



Ressources

About in cellulo labeling
About in vitro labeling

The molecular probes handbook

<https://www.thermofisher.com/fr/fr/home/references/molecular-probes-the-handbook.html>

About fluorescent proteins

<https://www.fpbases.org/>

Some books we can share....

